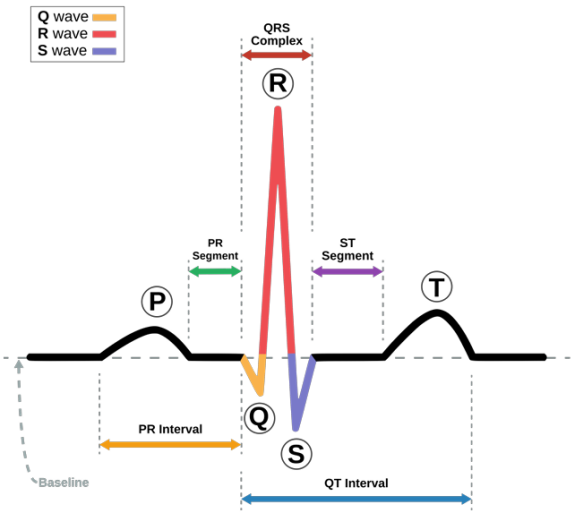
THE UNIVERSITY OF MELBOURNE

Final Report

The Art of Scientific Computing: Project Title: Respiratory Rates

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Chapter 1

Introduction

1.1 Background

The respiratory rate is an important measure of a person's health and diagnosis of illnesses, defined as the number of completed breath cycles taken per minute. [1][2] While respiratory rates can be affected by non-disease related factors such as exercise, emotion and elevation, in this study we are interested in Apnoea. Apnoea (commonly known as sleep apnea) is an increasingly common disorder characterised by repeated airway collapses during sleep, which can lead to oxygen desaturation and disrupted sleep. [3] However, the measurement of respiratory rates directly can be an invasive , expensive and at times an inaccurate process. This has led to the underdiagnosis of Apnoea, as the diagnosis of Apnoea requires respiratory measurements.
This has motivated research into deriving respiration from Electrocardiograms (ECGs), as these measurements are low-cost and non-invasive. [2] In this study, we will be investigating methods of deriving respitatory rates through ECGs.

1.2 Methodology

The language used was Python. The packages used were ... To analyse, we used (insert tecnique).

1.3 Results

1.4 Discussions and further direction

1.5 Computer code

1.5.1 Code input/output

1.5.2 Code written verbatim

This can be done with the `listings` and `verbatim` packages. You should modify the preamble document so that these packages correctly interpret the programming language that you used.

This will eventually be replaced with real code used for the project.

```
      FUNCTION PINEWT(N)
      IMPLICIT DOUBLE PRECISION (A-H,O-Z)
C*********************************************************************C
C     PINEWT EVALUATES THE INFINITE SERIES REPRESENTATION             C
C     FOR PI AS A TRUNCATED SUM.                                      C
C*********************************************************************C
C
C     ENSURE N IS A LEGAL VALUE
      IF(N.LT.0) THEN
        WRITE(6,*) 'In PINEWT: illegal value for N.',N
        RETURN
      ELSEIF(N.EQ.0) THEN
        PINEWT = 2.0D0
      ENDIF
C
C     INITIALISE SOME COUNTERS
      TWOPOW = 1.0D0
      FACKNM = 1.0D0
      FACKDM = 1.0D0
      PINEWT = TWOPOW*FACKNM*FACKNM/FACKDM
C
C     LOOP OVER SERIES ORDER N TIMES, ADDING EACH TERM TO TOTAL
      DO I=1,N
        TWOPOW = 2.0D0*TWOPOW
        FACKNM = DFLOAT(I)*FACKNM
        FACKDM = DFLOAT(2*I+1)*DFLOAT(2*I)*FACKDM
        PINEWT = PINEWT + TWOPOW*FACKNM*FACKNM/FACKDM
      ENDDO
C
C     DOUBLE THE RESULT (SEE WIKIPEDIA FORMULA)
      PINEWT = 2.0D0*PINEWT
C
      RETURN
      END
```

Appendix A

Program flowchart

A flowchart is a conceptual aid which highlights input and output that a program should have, as well as the consequences of any decisions that it must make. Sometimes, these will include *pseudocode*, which mimics some of the more common coding structures in a universal way. (For instance, you might say 'loop over all array addresses'.)

This particular flowchart indicates a program that reads three numbers from the terminal, and then produces the largest of the numbers back to the terminal. The colour scheme is not mandatory, but helps the reader to know what is going on at a glance.

This flowchart was drawn with the free 'Lucidcharts' extension to Google Docs, from which the result was exported as an `svg` file, and any unnecessary features were then cropped out with Inkscape. Choose any method you wish to create your own flowchart!

Bibliography

[1] A. L. Association. Understanding Vital Signs: The Importance of Your Respiratory Rate. https://www.lung.org/blog/respiratory-rate-vital-signs.

[2] J. Fan, S. Yang, J. Liu, Z. Zhu, J. Xiao, L. Chang, S. Lin, and J. Zhou. A High Accuracy & Ultra-Low Power ECG-Derived Respiration Estimation Processor for Wearable Respiration Monitoring Sensor. *Biosensors*, 12(8):665, Aug. 2022.

[3] A. S. Jordan, D. G. McSharry, and A. Malhotra. Adult obstructive sleep apnoea. *The Lancet*, 383(9918):736–747, Feb. 2014.